

Change and the Environment: A journey of statisticians and environmental scientists

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MARS

Mathematics for AI
in Real-world Systems

School of
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- £15M investment in applied mathematics and AI research
- 10 new academics
- Research Fellows
- RSE



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Engineering



- Developing methods for analyzing data whose statistical properties change over time.

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But what does that mean?

Interesting things?



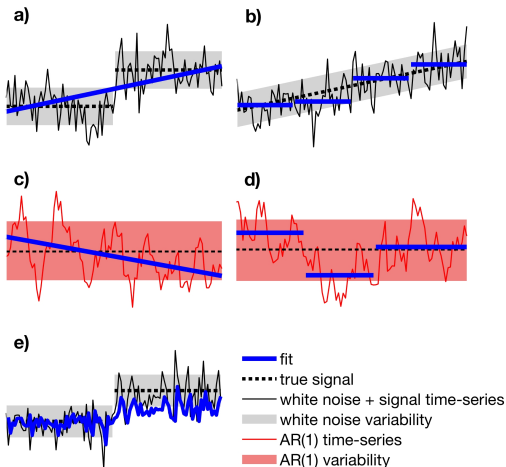
... in collaboration



over 140 co-authors over 13 years.



Challenges



- Temperatures
- River flows
- Flooding
- Soil (moisture and microbial)
- Trees
- Ice sheets
- Air quality
- Water (River and lake) quality
- Precipitation
- Carbon sequestration
- ...

Is climate change speeding up? Here's what the science says.

This year's record temperatures have some scientists concerned that the pace of warming may be accelerating. But not everyone agrees.

By [Chris Mooney](#) and [Shannon Osaka](#)

December 26, 2023 at 6:30 a.m. EST

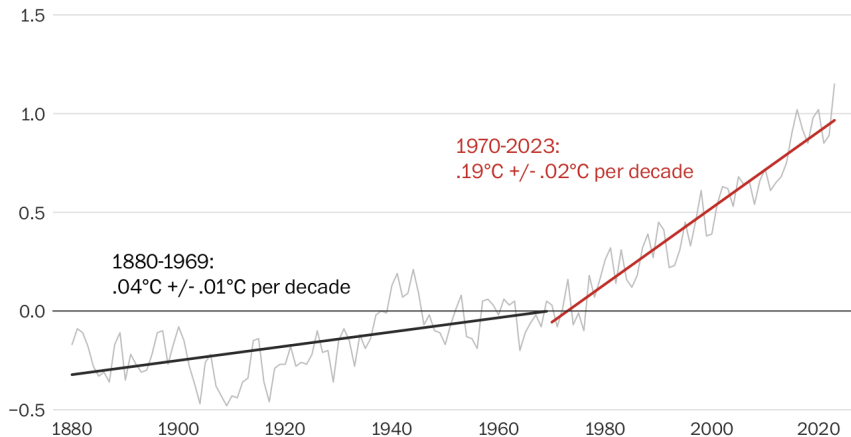
For the past several years, a small group of scientists has warned that sometime early this century, the rate of global warming — which has remained largely steady for decades — might accelerate. Temperatures could rise higher, faster. The drumbeat of weather disasters may become more insistent.

Where is Statistics?



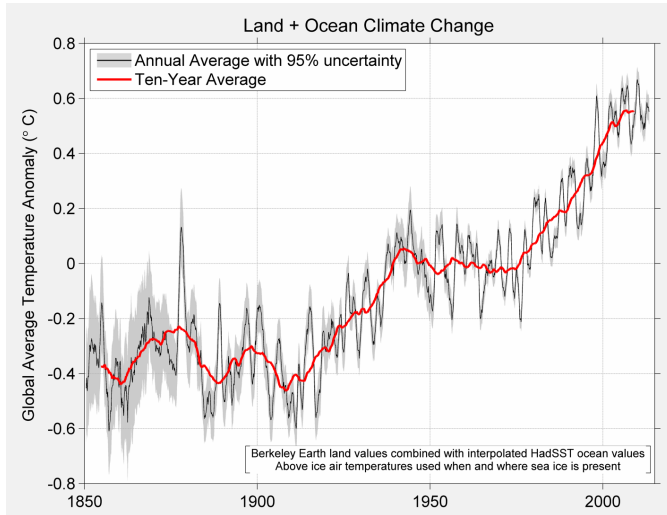
The increased rate of global warming

Values are relative to the 1951-1980 global mean temperature, in degrees Celsius

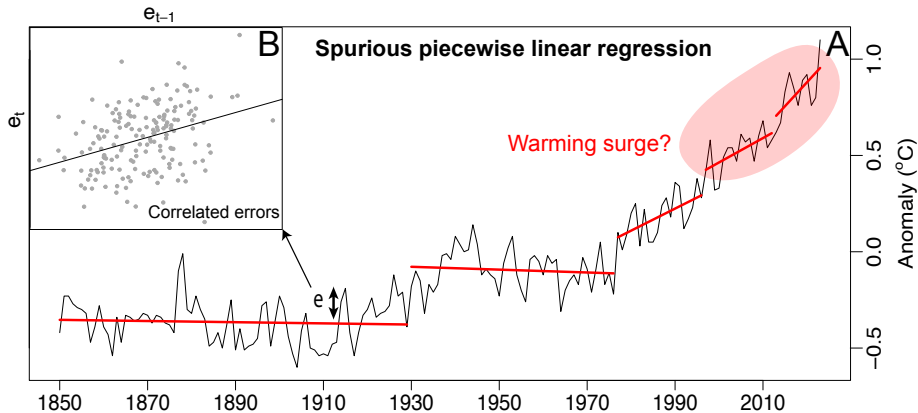


Note: 2023 is an estimate based on values from January through November.

Application: Temperature

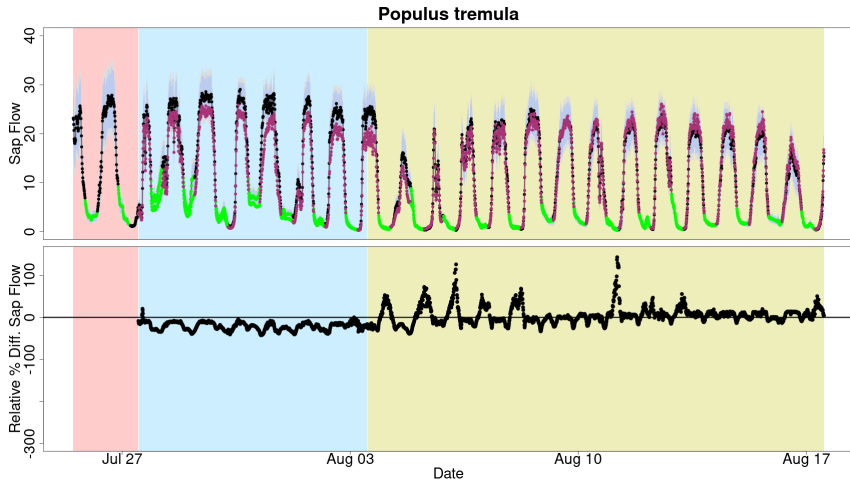


Beaulieu, Killick (2018) J.Climate



Beaulieu, Gallagher, Killick, Lund, Shi (2024+) Comm. Earth & Env.

Application: Sap Flux



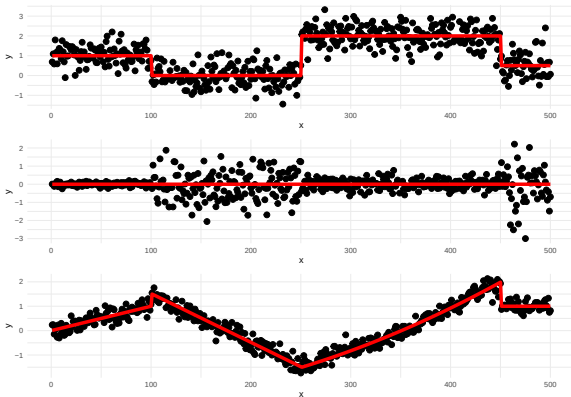
Hirons, Killick, Gong, Sjöman (2025+) In preparation

What are changepoints?



For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

There are many different types of change.



What are changepoints?

Changepoints are also known as:

- breakpoints
- segmentation
- structural breaks
- regime switching
- detecting disorder

and can be found in a wide range of literature including

- quality control
- finance
- medicine
- environment
- linguistics
- ...

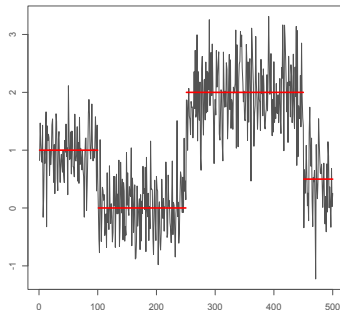
Example



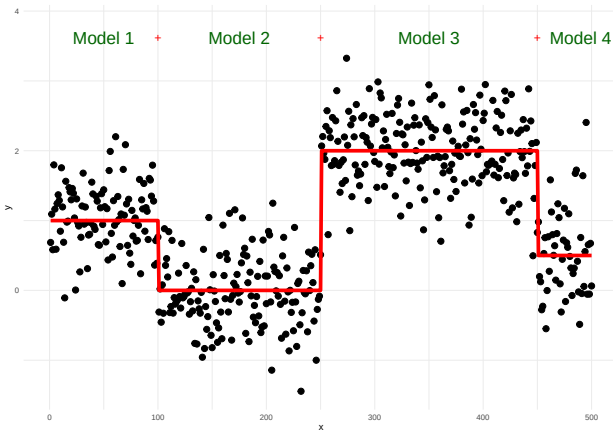
Assume we have time-series data
where

$$Y_t | \theta_t \sim N(\theta_t, 1),$$

but where the means, θ_t , are
piecewise constant through time.



Fitting changepoints



- How many changes?
- Where are the changes? 2^{n-1} possible solutions!

If we define

$$\mathcal{P}_t = \{\tau : 0 < \tau_1 < \dots < \tau_m < t\}$$

$$F(t) = \min_{\tau \in \mathcal{P}_t} \left\{ \sum_{i=1}^{m+1} C(y_{(\tau_{i-1}+1):\tau_i}) + \beta \right\}$$

So,

$$F(t) = \min_{\tau^*} \left\{ \min_{\tau \in \mathcal{P}_{\tau^*}} \left[\sum_{i=1}^m C(y_{(\tau_{i-1}+1):\tau_i}) + \beta \right] + C(y_{(\tau^*+1):t}) + \beta \right\}$$

Thus we minimise,

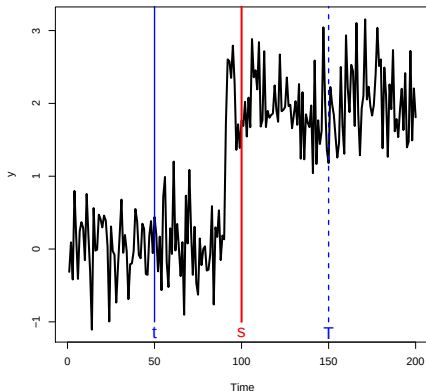
$$F(t) = \min_{\tau^*} \{ F(\tau^*) + \mathcal{C}(y_{(\tau^*+1):t}) + \beta \}$$

Recursively solving the minimisation for $t = 1, \dots, n$ gives an algorithm that is $\mathcal{O}(n^2)$.



The condition in the theorem just means that for any $T > s$ the best partition which involves a changepoint at s will be better than one which have $[t, T]$ as a single segment.

Thus t can never be the (optimal) most recent changepoint prior to T for all $T > s$.



Killick, Fearnhead, Eckley (2012) JASA

If many t are pruned, excluded from the minimisation then computational time will be drastically reduced.

We can prove that, under certain regularity conditions, the expected computational complexity will be $\mathcal{O}(n)$.

The most important condition is that *the number of changepoints increases linearly with n .*

This is natural in many applications: e.g. as you collect time-series data over larger time-periods; or genomic data over larger regions of the genome.

$$F(t) = \min_{\tau^*} \{ F(\tau^*) + \mathcal{C}(y_{(\tau^*+1):t}) + \beta \}$$

- $\mathcal{C}(\cdot)$ is very general
- Need independence of parameters across segments (can have AR(1) parameter generalization)
- Need that adding a changepoint to a segment decreases the cost

Thus becomes all about model choice (again!).



- potentially hundreds or thousands of series
- no time to consider the format of change for each
- need to include both the potential for trends and also red noise (autocorrelation).

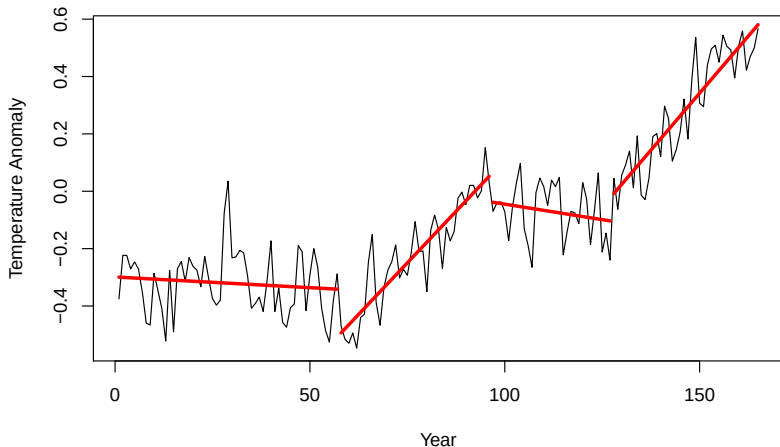
Use `EnvCpt` package on CRAN!

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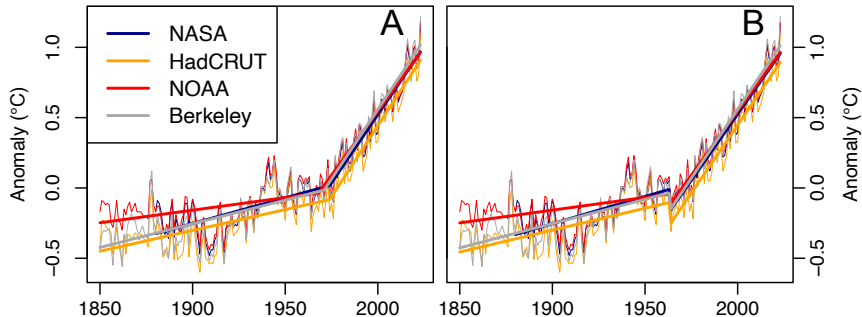
- Debate over whether trends should be continuous or discontinuous
- Debate over whether autocorrelation is there and whether changes over time

Application: Temperature



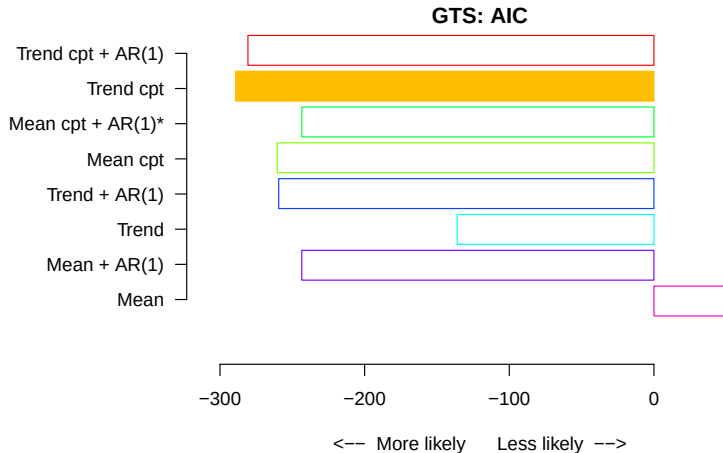
Beaulieu, Killick (2018) J. Climate

Application: Temperature



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Application: Temperatures

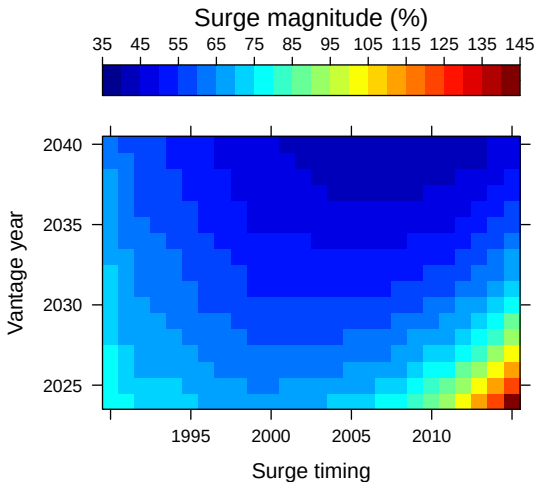


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So what?



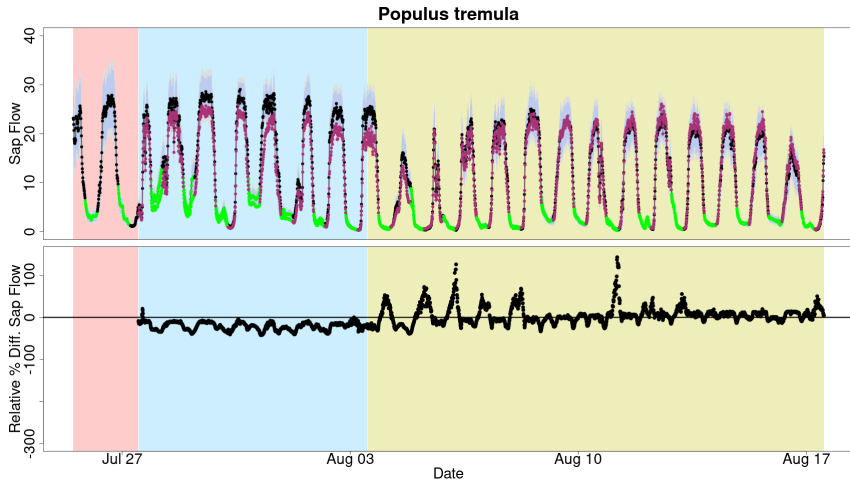
What would we need to see (in the statistically preferred model)?



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- Tolerance of trees is important for planning
 - Trees are vital to climate targets
 - Changes to weather mean more droughts and/or local flooding
-
- Waterlogged trees for two weeks
 - Monitored two weeks after water was drained
 - Looking at difference across species

Application: Sap Flux

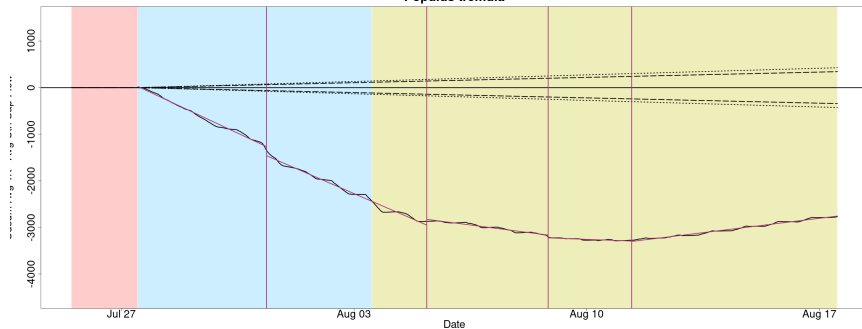


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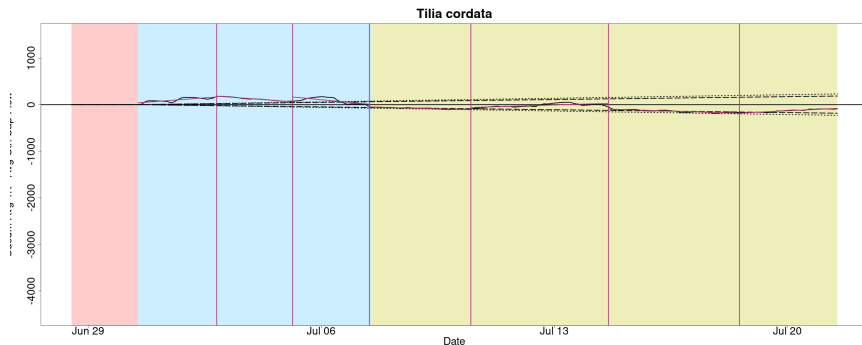
Difference CUSUM control chart with offline changepoint trends

Populus tremula





Difference CUSUM control chart with offline changepoint trends



Appropriate maths available $\neq$ Appropriate maths used

What helps:

- Available software
- ... that is easy to use
- ... has good documentation
- ... and is “findable”
- maintainers who are willing to answer basic queries

- Time series and changepoints are an important part of environmental applications
- Newer sensors are producing more time series data which can pose a challenge
- Collaborations are key, but should be mutually beneficial
- Lots of environmental applications and challenges to work on

All results from this talk are available via reproducible scripts. R packages used: EnvCpt (CRAN), changepoint (CRAN), changepoint.forecast (Github).